

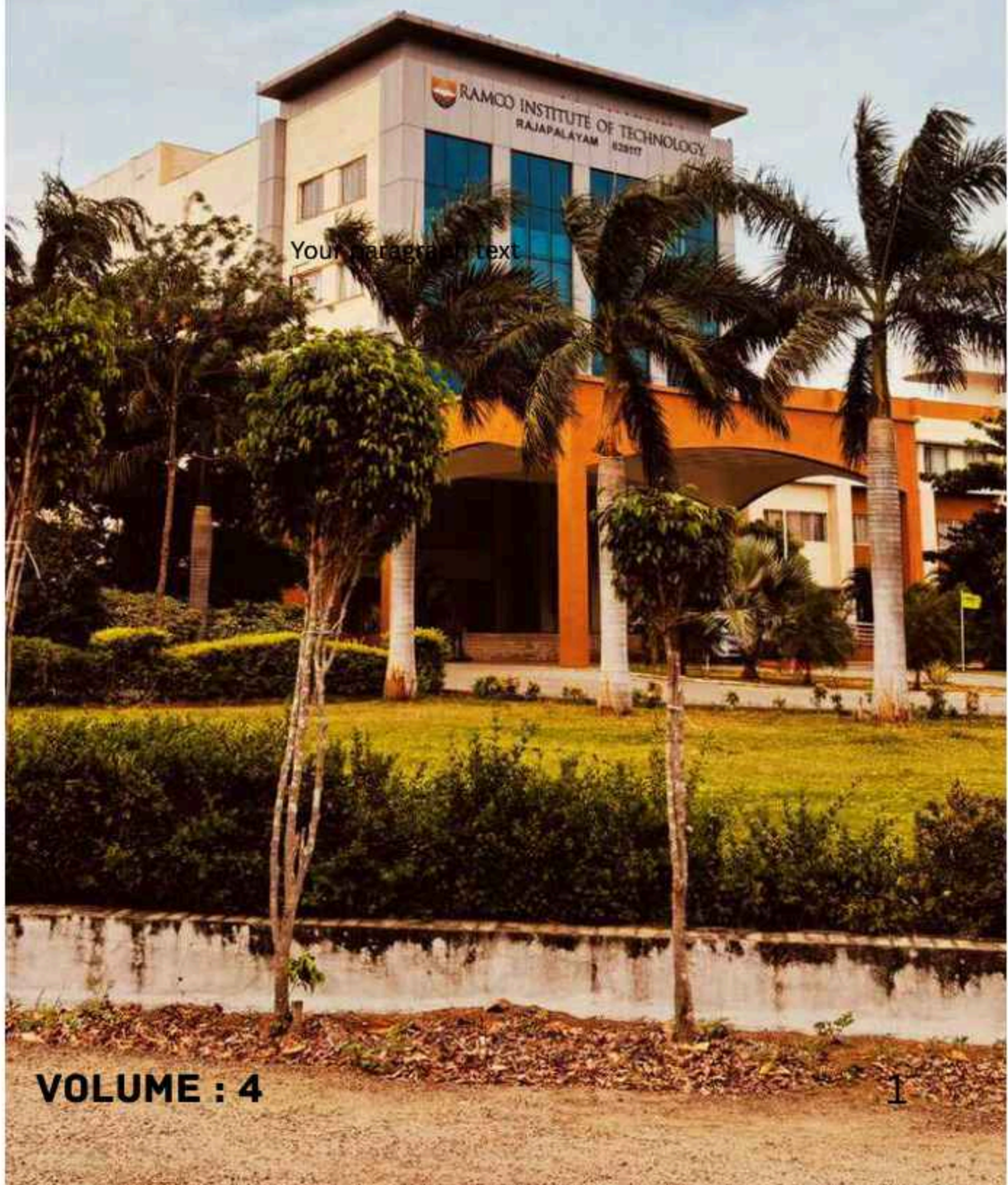


RAMCO
INSTITUTE OF
TECHNOLOGY

2023-2024

DEPARTMENT OF CIVIL ENGINEERING

STRUCTA



VOLUME : 4

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ABOUT THE DEPARTMENT

The Department of Civil Engineering was established in the year 2013. At present, it offers undergraduate programme, B.E. in Civil Engineering with a sanctioned intake of 60 students. The Department broadly covers important fields such as Surveying, Geotechnical Engineering, Structural Engineering, Hydraulics and Water Resources Engineering, Construction Materials and Practices, Environmental Engineering and Transportation Engineering.

The Department has well established modern laboratories to cater to the needs of academic and research activities. It is well equipped with computational facilities and resources both in terms of hardware and software. The Department constantly facilitates students from other departments of the Institute as a gesture of support to their academic, co-curricular and research desires.

The Department has well-qualified and committed faculty members, lab technicians and modern infrastructure in the form of elegant and smart classrooms, spacious laboratories, latest machinery and equipment. Students are encouraged to participate in co-curricular and extra-curricular activities to acquire leadership qualities. The Civil Engineering Association RIT STRUCTA has been functioning for the benefit of the students. It serves as a platform for students to make technical presentations, gain knowledge about recent advancements and bring together novel ideas. This Association aims at bridging the gap between the academic and the industry by inviting renowned industrialists and academicians to interact with students.

Guest lectures, industrial visits, site visits, etc., are organized and value added courses are conducted on regular basis to ensure that students are conversant with recent developments in the field of Civil Engineering.

The Department encompasses professional societies such as Indian Concrete Institute (ICI), ISTE chapters, which provides students a platform to acquire knowledge in various technical activities. It offers consultancy services in the fields of Structural Engineering, Geotechnical Engineering, Environmental Engineering, Surveying and Highway Engineering.

Vision

Create knowledgeable and professionally competent Civil Engineers who are capable of providing global solutions to the problems related to Civil Engineering.

Mission

- Impart high quality education and technical skills to transform students into Professional Civil Engineers.
- Establish state-of-the-art infrastructure for updating knowledge in recent developments and cutting edge technology.
- Bridge the gap between academics and industry by practical training and providing solutions for design projects related to Civil Engineering.
- Motivate the faculty members and students in the teaching learning process for achieving excellence to meet the global standards.
- Produce graduates with high professional and ethical standards for fulfilling the needs of the industry and society.

DEPARTMENT OF CIVIL ENGINEERING

Program Educational Objectives (PEOs)

- Graduates of the programme will be professionally competent and apply the fundamental concepts of mathematics, science and Civil Engineering to solve problems in Civil Engineering.
- Graduates of the programme will perform their responsibilities as professional Civil Engineers with quality and ethics.
- Graduates of the programme, as part of an organization or as Entrepreneurs, will continue to learn with evolving technologies.
- Graduates of the programme will innovate and provide global solutions for a sustainable built environment.
- Graduates of the programme will contribute to the development of sustainable Infrastructure for the betterment of society.

Program Specific Outcomes (PSOs)

On completion of the degree, the students will be able to:

- Communicate and present civil engineering projects effectively.
- Use the techniques, skills, and modern engineering tools necessary for civil engineering practice and project management.
- Provide sustainable solutions to civil engineering problems.
- Perform as design consultants in construction industry for the design of civil engineering structures.



Program Outcomes (POs)

Engineering Graduates will be able to:

- **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- **Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to ones own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

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*POP
GOES
MY
ART*

WE ARE THE MASTERS
OF OUR IMAGINATION

Art.

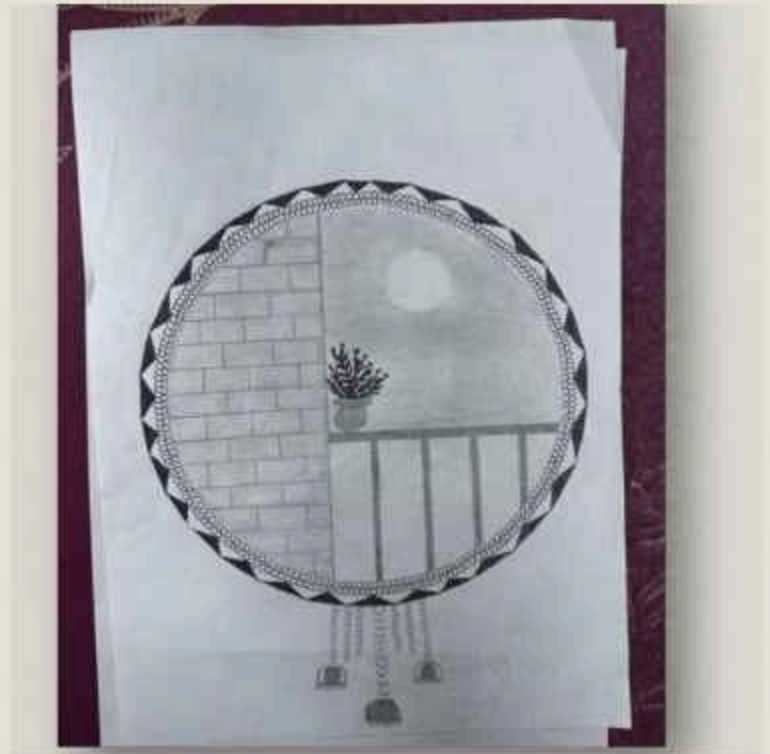
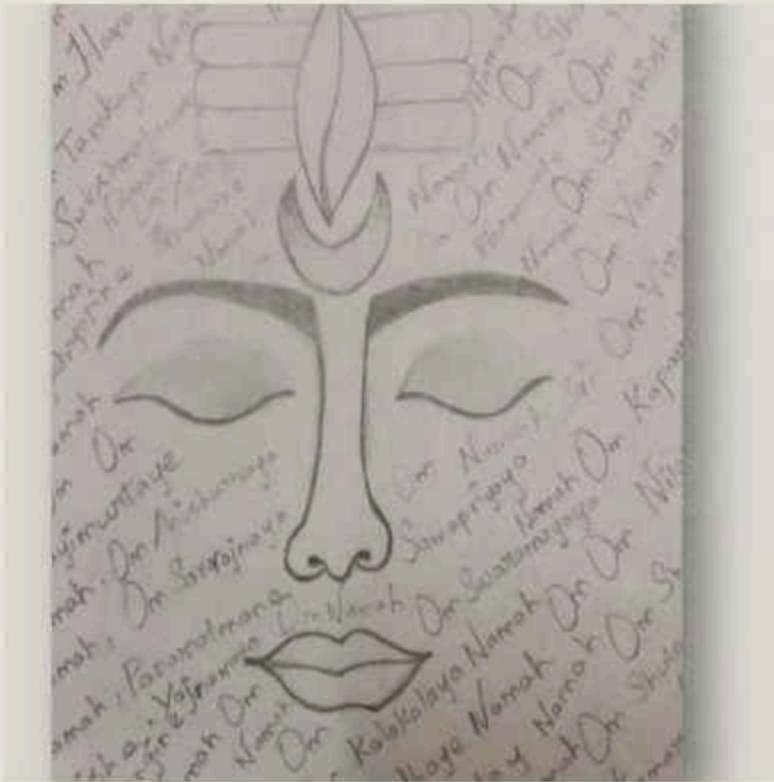
The Art

Mr. A Velprasath
1st year



2023-2024

ARTIST *Portfolio*



Mr. G Ari kara Moorthy
II Civil

The Cat

IS A LOCAL STREET ART MAGAZINE
IN MANILA THAT DISCUSSES
CURRENT OR RECENT NEWS OF
EITHER GENERAL INTEREST OR OF A
SPECIFIC TOPIC WITHIN THE METRO.



THE ART

MR. M Kavikumar

II CIVIL

The Cat

ARTIST

Mr. Muruga sundar M
I Civil



ART MAGAZINE

The leading magazine for artists,
photographers, and graphic professionals



THE GREEN BUILDING

THE ARTICLES

ENGINEERING

MAGAZINE

Ratan Tata

AN ICON OF INDIAN INDUSTRY



Ratan Tata, the former Chairman of Tata Sons and one of India's most respected business leaders, is renowned for his visionary leadership and ethical approach to business. Under his stewardship, the Tata Group transformed from a predominantly Indian business into a global conglomerate, with interests spanning across steel, automobiles, IT services, telecommunications, and hospitality. His leadership style is characterized by humility, a commitment to values, and a passion for innovation. Here's a deeper look into the leadership qualities and principles that define Ratan Tata



Ratan Tata's Inspiring Quotes

1. I don't believe in taking right decisions. I take decisions and then make them right.
2. Ups and downs in life are very important to keep us going, because a straight line even in an ECG means we are not alive.
3. If you want to walk fast, walk alone. But if you want to walk far, walk together.
4. None can destroy iron, but its own rust can. Likewise, none can destroy a person, but their own mindset can.
5. Power and wealth are not two of my main stakes.
6. I have always been very confident and very upbeat about the future potential of India. I think it is a great country with great potential.
7. Business needs to go beyond the interest of their companies to the communities they serve."
8. Take the stones people throw at you, and use them to build a monument.
9. People still believe what they read is necessarily the truth.

How the National Highways are Numbered????

Can you know, how the National Highways are numbered in our country? If yes, good, otherwise, no issues! In this article, we will explore how the National highways numbered in our country. Can you tell me which is the longest highway in our country running from Kashmir to Kanyakumari? Most of us will say, it is NH7. That's an old number. Some years back, the Ministry of Road Transport and Highways rationalized the numbering of highways, and the same was published in the Gazette on 28th April 2010. Let's delve into the rationalization process of National Highways in our country.

It is a systematic numbering scheme based on the highway's orientation and geographic location.

For all the national highways from North to South, even numbers are used and it is assigned from East to West in the ascending order. In other words, for high longitudes, lower numbers, and lower longitudes high numbers are assigned. NH-2 is located in North Eastern States and NH-68 is located in Rajasthan.

For all the highways from East to West, odd numbers are assigned. In this case, the highways are named in ascending numbers from East to West. In other words, for high latitudes, lower numbers, and for lower latitudes higher numbers are assigned. For example, NH-8 is located in J&K, and NH-87 is located in Tamil Nadu.

If we move from East to West, the numbers used for highways located in the North to South direction increases. For example, if the number four is used for a highway located on North-South but in an Eastern State, then the number for a highway located on North to South but in a Central State or Western State will always be denoted by more than four.

Highways that are denoted by three-digit numbers are known as subsidiary highways. For Example, 244, 144, and 344 are the branches of National Highway number 44. In the subsidiary highway number, if the first digit is odd, then the highway is located from East to West and if the first digit is even, it is located from North to South.

References:

1. https://en.wikipedia.org/wiki/2010_renumbering_of_national_highways_in_India#:~:text=On%2028%20April%202010%2C%20the,geographic%20location%20of%20the%20highway.
2. <https://www.jagranjosh.com/general-knowledge/how-national-highways-are-named-in-india-1484805172-1>

MR.T.CHOCKALINGAM
AP(SG)/CIVIL

BIOLOGICAL CONCRETE

Dr.M.Indhumathi ASP/Civil

Biological concrete, also known as bio-concrete, is a sustainable construction material that integrates biological processes into its composition. The key feature of bio-concrete is its ability to heal itself when cracks form. This is achieved by embedding specific bacteria within the concrete mix that can precipitate calcium carbonate when exposed to moisture.

Here are some notable aspects of biological concrete:

1. **Self-Healing Properties:** When cracks occur, the bacteria become active, consume nutrients, and produce calcium carbonate, which fills the cracks and restores the integrity of the structure.

2. **Sustainability:** By reducing the need for repairs and extending the lifespan of concrete structures, bio-concrete can lower the overall environmental impact associated with construction and maintenance.

3. **Materials:** The production of bio-concrete may involve using waste materials or alternative aggregates, further enhancing its eco-friendliness.

Overall, biological concrete represents a promising innovation in the field of sustainable construction, aligning with efforts to reduce the carbon footprint of building materials. Biological concrete has several innovative applications across various construction projects, leveraging its self-healing properties and sustainability features. Here are some detailed applications:

1. **Marine Structures:** In coastal and underwater applications, bio-concrete can withstand harsh environments and saltwater conditions. Its self-healing ability helps maintain the integrity of structures like piers, seawalls, and marine habitats.

2. **Historical Restoration:** Bio-concrete can be employed in the restoration of historical buildings. Its compatibility with traditional materials allows for repairs that maintain aesthetic and structural integrity while enhancing durability.

1. **Infrastructure Repair:** Bio-concrete is particularly useful for repairing aging infrastructure such as bridges, highways, and tunnels. The self-healing mechanism can significantly reduce maintenance costs and downtime, allowing structures to remain operational longer without extensive repairs.

2. **Building Construction:** In residential and commercial buildings, bio-concrete can be used in walls, floors, and foundations. Its ability to heal minor cracks can improve the longevity and durability of the structure, reducing the need for frequent maintenance.

3. **Water Retaining Structures:** Bio-concrete is ideal for applications like water tanks, reservoirs, and sewage systems. Its self-healing properties can help prevent leaks and water loss, ensuring efficient operation and lower maintenance needs.

4. **Pavements and Roads:** Incorporating bio-concrete in road construction can enhance the durability of pavements. As vehicles cause micro-cracks over time, the self-healing bacteria can activate and repair these cracks, maintaining the road surface and reducing repair frequency.

5. **Landscaping and Green Spaces:** Bio-concrete can be utilized in parks, green roofs, and landscape features. It supports plant growth while providing structural integrity, and its self-healing properties contribute to the sustainability of outdoor spaces.

6. **Innovative Architectural Designs:** Architects can explore creative designs with bio-concrete, knowing that its self-healing properties can help maintain the integrity of intricate shapes and structures over time.

7. **By integrating biological concrete into these applications, the construction industry can move towards more sustainable practices while enhancing the durability and lifespan of built environments.**



EDIBLE WATER BUBBLE & AIR WATER GENERATOR

Mrs.C.Subha AP(SG)/Civil

- ü The edible water bubble, or "Ooho", is a biodegradable water container made from seaweed extract.

- üThe edible water sphere is made using spherification, where water is encapsulated in a thin, edible membrane made from sodium alginate (derived from seaweed) and calcium chloride.

- ü It offers an eco-friendly alternative to plastic bottles, reducing plastic waste.

- ü The outer membrane is completely edible and decomposes naturally if not consumed. Each bubble holds a small amount of water, making it ideal for single-use hydration during events or sports.

- üThis innovation has gained attention for its potential to combat the growing problem of environmental plastic pollution.



Atmospheric Water Generation (AWG) is a technology that extracts water from the air by condensing moisture to produce drinkable water. This process leverages the humidity in the atmosphere, using either cooling-based condensation (by cooling air below its dew point) or desiccant-based absorption (using materials that absorb moisture from the air). AWG systems are designed to provide clean water in areas with limited freshwater sources but sufficient humidity, making it a promising technology for water-scarce regions and emergencies.



3D PRINTING IN CONSTRUCTION

introduction: The field of civil engineering construction is witnessing a transformative shift with the introduction of 3D printing technology. Also known as additive manufacturing, 3D printing is revolutionizing the way buildings and infrastructure are designed and constructed. This article explores the applications and benefits of 3D printing in civil engineering, highlighting its potential to enhance efficiency, sustainability, and creativity in construction projects.

1. Faster and Cost-Effective Construction: Traditional construction methods often involve time-consuming and labor-intensive processes. 3D printing enables rapid prototyping and fabrication of complex structural components, significantly reducing construction time. By eliminating the need for formwork and manual assembly, 3D printing can lead to substantial cost savings in terms of labor, materials, and logistics.

2. Design Freedom and Customization: 3D printing allows for intricate and customizable designs that were previously unattainable with conventional construction methods. Complex geometries, organic shapes, and intricate architectural details can be seamlessly translated into physical structures. This design freedom opens up new possibilities for architects and engineers to create innovative and visually striking buildings while ensuring structural integrity.

3. Resource Efficiency and Waste Reduction: Sustainable construction practices are increasingly vital in today's world. 3D printing offers inherent resource efficiency by using only the necessary amount of material required for construction. By minimizing waste during the printing process, this technology helps reduce the environmental impact of construction projects. Additionally, the use of recycled or locally sourced materials can further enhance the sustainability of 3D-printed structures.

4. Enhanced Structural Performance: The layer-by-layer construction process of 3D printing allows for the integration of structural reinforcements and optimized load-bearing capabilities. This results in stronger, more resilient structures that can withstand extreme weather events and seismic activities. The ability to precisely control the material composition and density during printing enables the creation of lightweight yet robust structures, enhancing overall structural performance.

TITLE

REVOLUTIONIZING CIVIL ENGINEERING CONSTRUCTION: THE POWER OF 3D PRINTING

Mrs.A.Leema Margret
AP/Civil



3D PRINTING IN CONSTRUCTION

TITLE

REVOLUTIONIZING CIVIL ENGINEERING CONSTRUCTION: THE POWER OF 3D PRINTING

Mrs.A.Leema Margret AP/Civil

CIVIL MINT

5. **Construction in Challenging Environments:** 3D printing technology offers unique advantages for construction in remote or challenging environments. By reducing the reliance on conventional construction materials and equipment, 3D printers can be transported to hard-to-reach areas, enabling the construction of temporary shelters, disaster relief housing, or infrastructure in areas with limited access to resources and skilled labor.

6. **Repairs, Maintenance, and Retrofitting:** Beyond new construction, 3D printing has applications in repairs, maintenance, and retrofitting of existing infrastructure. With the ability to quickly produce customized replacement parts, 3D printing can streamline the repair process and prolong the lifespan of structures. It also enables the integration of sensors and smart components, facilitating real-time monitoring and predictive maintenance of infrastructure assets.

7. **Collaboration and Innovation:** The advent of 3D printing in civil engineering construction encourages collaboration and interdisciplinary approaches. Architects, engineers, material scientists, and computer programmers can work together to optimize designs, develop new materials, and refine the printing process. This collaboration fosters innovation and pushes the boundaries of what is achievable in the realm of construction.

Conclusion: 3D printing is transforming the field of civil engineering construction, offering unprecedented opportunities for efficiency, sustainability, and design creativity. The ability to rapidly fabricate complex structures, reduce waste, enhance structural performance, and address challenges in diverse environments makes 3D printing a game-changer. As the technology continues to evolve and gain wider acceptance, we can expect a paradigm shift in the construction industry, where 3D printing plays a pivotal role in shaping the cities and infrastructure of the future.



As urbanization accelerates and populations grow, cities around the world face increasing challenges related to traffic congestion, air pollution, and road safety. Traditional traffic management systems are often insufficient to address these complex issues, prompting the need for smarter, more efficient solutions. Smart traffic management systems leverage advanced technologies, including the Internet of Things (IoT), artificial intelligence (AI), and real-time data analytics, to optimize traffic flow and improve transportation efficiency.

Singapore: Smart Mobility System

- *Technology Used:* Singapore is a global leader in smart traffic management. Its system integrates sensors, GPS, and AI to optimize traffic flow and reduce congestion. The Expressway Monitoring and Advisory System (EMAS) detects accidents or breakdowns in real-time, sending alerts to traffic management centers for immediate response. Additionally, the Electronic Road Pricing (ERP) system dynamically adjusts toll prices based on traffic conditions, reducing traffic during peak hours.

[Institution of Civil Engineers \(ICE\)](#)

- *Impact:* Singapore's smart traffic system has led to a significant reduction in congestion, improved road safety, and enhanced public transit reliability. The system's adaptability ensures smooth traffic even in high-density urban areas.

Los Angeles, USA: Adaptive Traffic Control System (ATCS)

- *Technology Used:* Los Angeles implemented the Automated Traffic Surveillance and Control (ATSAC) system, which monitors traffic conditions in real-time using sensors and cameras at intersections. The system adjusts traffic light timings based on real-time traffic patterns, optimizing flow on major roads.
- *Impact:* ATSAC has improved travel times by up to 12% and reduced fuel consumption and emissions by limiting stop-and-go driving. This system covers over 4,500 traffic signals across the city, making it one of the largest of its kind globally.

Barcelona, Spain: Smart Traffic Management

- *Technology Used:* Barcelona's smart traffic system integrates IoT sensors, cameras, and data analytics to monitor traffic in real time. The city uses smart signals and predictive analytics to optimize traffic flows and reduce congestion. Additionally, Barcelona implemented smart parking solutions, allowing drivers to locate available parking spaces using a mobile app.
- *Impact:* The system has reduced congestion, improved air quality, and enhanced the efficiency of public transportation. The smart parking system alone has significantly reduced the time spent searching for parking, reducing emissions from idling vehicles.

Copenhagen, Denmark: Green Wave for Cyclists

- *Technology Used:* Copenhagen has implemented a unique smart traffic solution called the Green Wave, designed to prioritize bicycles over cars. Sensors detect the flow of bicycles during peak hours and adjust traffic lights to ensure that cyclists can move without stopping at multiple lights. This system is a part of Copenhagen's broader plan to become carbon-neutral by 2025.
- *Impact:* The Green Wave has increased cycling speeds and improved safety for cyclists, encouraging more people to use bicycles for commuting. This, in turn, has reduced car traffic and emissions
- *Smart traffic systems in cities like Singapore, Los Angeles, Barcelona, and Copenhagen are transforming urban mobility by reducing congestion, improving road safety, and encouraging the use of sustainable transportation. These cities leverage real-time data, adaptive systems, and advanced technologies like IoT and AI to enhance overall transportation efficiency and sustainability.*

Mrs.R.Kalaimani
AP/Civil



Building information modeling (BIM) is a process supported by various tools, technologies, and contracts involving the generation and management of digital representations of physical and functional characteristics of places. This article discusses the diverse career opportunities in Building Information Modeling for engineers and architects.

Building information models are data that can be extracted, transferred, or networked to support decision-making regarding a building or other built asset.

Career Options in BIM for Civil Engineers and Architects

Individuals use current BIM software, businesses, and government agencies who plan, design, construct, operate and maintain diverse physical infrastructures, such as water, refuse, electricity, gas, communication utilities, roads, railways, bridges, ports, and tunnels.

Building Information Modeling is an exceptional 3D model-based process that gives Architecture, Engineering, and Construction (AEC) professionals the insight and tools to plan, design, construct and manage buildings and infrastructure more efficiently.

Architecture: Make better design decisions, improve building performance, and collaborate more effectively throughout the project lifecycle.

Civil: Use intelligent, connected workflows to help improve predictability, productivity, and profitability

Construction: Digitize your construction site and connect project information from design through construction and handover.

MEP: Improve MEP (mechanical, electrical, and plumbing) design quality and collaborate in real-time to support the project delivery process.

Plant: Manage the design and construction of intelligent piping, structures, and processes more efficiently and collaboratively throughout the project life cycle and hand-off.

Structures: Explore how structural design and detailing software helps you win new business and support project delivery.

It is the latest game-changer as a technology solution. But like any new technology, there is a great need for a skilled workforce in this area. As the demand for BIM professionals goes up across the Architecture, Engineering, and Construction (AEC) industry, what are the long-term prospects of such a career?

Naturally, several young students of civil engineering and construction disciplines worldwide are interested in making a career.

Mrs.D.Darling Helen Lydia

AP/Civil

22

EFFECT OF ADDITION OF CARBIDE IN SEWAGE DEGRADATION ABSTRACT



INTRODUCTION:

In recent years, the use of toilet additives to treat faecal sludge in domestic sanitation facilities has received increasing attention partly due to large sludge generation and stringent environmental management regulations. Domestic faecal sludge is raw or partially digested slurry or semisolid usually generated from on-site toilet systems. Major faecal sludge problems include finding safe handling and appropriate disposal options especially in developing countries. Untreated faecal sludge is laden with pathogens which could cause illnesses such as diarrhoea, cholera and dysentery. Safe disposal of faecal sludge into the environment will require treatments to sanitise (kill pathogens) and also to stabilise faecal matter (reduce vector attraction).

This study investigated the effects of two mostly improvised chemical additives, calcium carbide and lambda super 2.5 EC (LSEC), on the physico-chemical and microbial characteristics of faecal sludge from toilets. The quality of faecal sludge was assessed before and after application of the chemical additives in an experimental setup of ten different treatment units including a control, and treatment replicates. The initial characteristic of the faecal sludge was slightly acidic with high content of slowly degradable organic matter. The experimental control without additives after 30 days showed reduction in BOD₅, COD, helminth eggs and sludge mass by a maximum of 30%, 34.7%, 99.8% and 55% respectively. Similarly, calcium carbide additive reduced the BOD₅, COD, helminth eggs and the mass of the faecal sludge by 47.4%, 48.3%, 99.6% and 61% respectively. Also, LSEC additive reduced BOD₅, COD, helminth eggs and the mass of the sludge by 40.6%, 47.9%, 95.9% and 58% respectively. The two additives showed significant treatment effect on the faecal sludge although the level of treatment could not meet the regulatory discharge limits for the key quality parameters assessed including sanitisation. The study is still a grey area and more research is recommended to enrich the findings.

MATERIAL AND REQUIREMENTS:

Faecal sludge sources and sample collection:

Samples of faecal sludge (FS) were taken from the only three (3) functional public toilets at the Ayeduase and Kotei communities, which are all suburbs of the Oforikrom Municipal Assembly in Kumasi, Ghana. The two communities are neighbours to the Kwame Nkrumah University of Science and Technology (KNUST), the institution where the experimentation was done. Ayeduase and Kotei are known as dormitory towns because of their high student population. The majority (70%) of the inhabitants are served by on-site sanitation systems such as public toilets with technologies like Kumasi ventilated improved pits (KVIPs), and pour-flush and cistern flush toilets connected to septic tanks. One of the key challenges faced by managers of these public sanitation facilities is cost burden associated with the management of faecal sludge generated. The toilet facilities receive high usage rates, with quickly filled up pits and always requiring frequent desludging incurring higher cost. The challenging situation was same for the only functional toilets from which faecal sludge samples were taken at the time of the study. The three (3) toilets were desludged within short periods of every

Experimental setup and laboratory analyses:

The experimental setup and laboratory analyses followed standard procedure described by Foxon et al., The main additives tested in this study were calcium carbide and lambda super 2.5 EC (LSEC).

CONCLUSION:

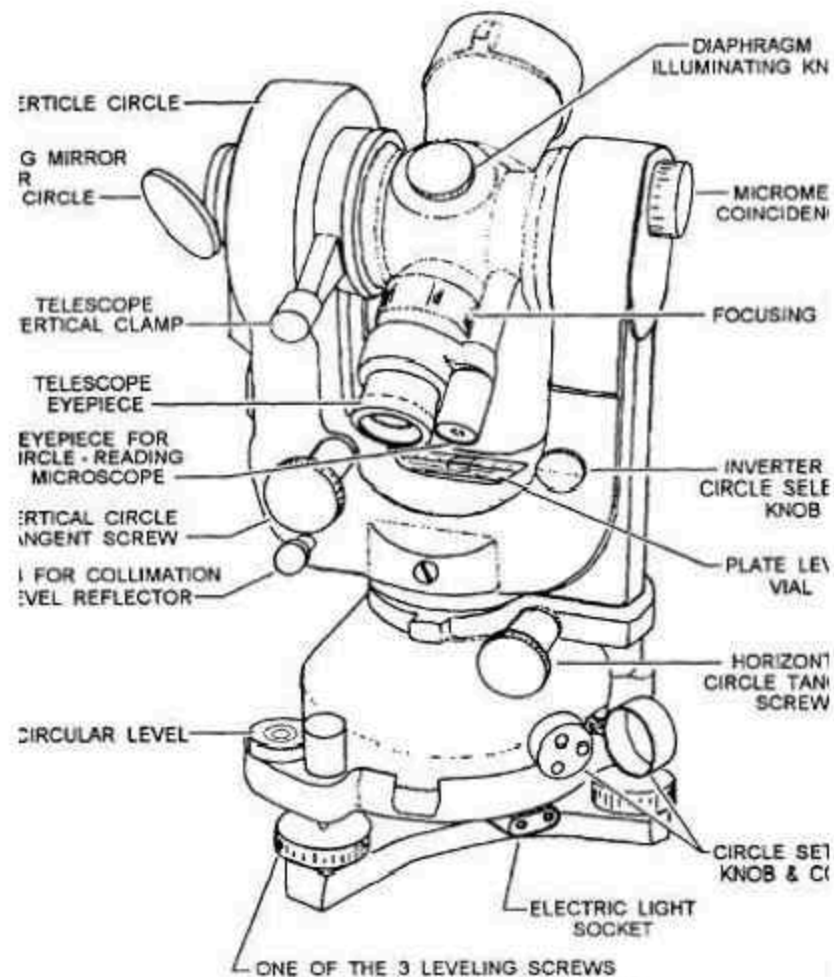
The study showed that faecal sludge from the public toilets was slightly acidic with high contents of moisture and microbial loads as expected, and also high levels slowly degradable organic matter, thus more COD than BOD5 giving it a lower biodegradability property. The two additives tested especially calcium carbide showed some influence on the faecal sludge physico-chemical properties like pH and temperature increase or decrease immediately at the early stages of treatment but stabilized shortly afterwards to ambient conditions. Statistically, the additives could significantly treat the faecal sludge to appreciable levels by moisture content reduction between 39 – 74%, BOD5 (30–47%), total coliform (79–98.5%) and helminth eggs

Ms. N.Sudha

II Civil

ADVANCE TECHNOLOGY IN SURVEYING

- Definition of Surveying:
- Surveying has to do with the determination of the relative spatial location of points on or near the surface of the earth.
- It is the art of measuring horizontal and vertical distances between objects, of measuring angles between lines, of determining the direction of lines, and of establishing points by predetermined angular and linear measurements.
- Along with the actual survey measurements are the mathematical calculations. Distances, angles, directions, locations, elevations, areas, and volumes are thus determined from the data of the survey.
- Survey data is portrayed graphically by the construction of maps, profiles, cross sections, and diagrams.
- The importance of the Surveying:
- Land surveying is basically an art and science of mapping and measuring land. The entire scope of profession is wide; it actually boils down to calculate where the land boundaries are situated. This is very important as without this service, there would not have been railroads, skyscrapers
- could not have been erected and neither any individual could have put fences around their yards for not intruding others land.



THEODOLITE :

a precision instrument for measuring angles in the horizontal and vertical planes. Theodolites are used mainly for surveying applications, and have been adapted for specialized purposes such as meteorology and rocket launch. The theodolite is a complex instrument used mainly for accurate measurement of horizontal and vertical angle up to 10" or 20" depending upon the least count of the instrument. Because of its various uses, the theodolite is sometimes known as "Universal Instrument". Uses of theodolite: Following are the different purpose for which theodolite can be used

- 1) Measuring horizontal angle
- 2) Measuring vertical angle
- 3) Measuring deflection angle
- 4) Measuring magnetic bearing
- 5) Measuring the horizontal distance between two points
- 6) Finding vertical height of an object
- 7) Finding difference of elevation between various points
- 8) Ranging of a line

Different parts of theodolite:

EDM

Measurement of distance is accomplished with a modulated microwave or infrared carrier signal, generated by a small solid-state emitter within the instrument's optical path, and bounced off of the object to be measured. The modulation pattern in the returning signal is read and interpreted by the onboard computer in the EDM. The distance is determined by emitting and receiving multiple frequencies, and determining the integer number of wavelengths to the target for each frequency. In EDM the beam of light is the carrier and which is reflected



GEODETIC SURVEYING:

The type of surveying that takes into account the true shape of the earth. These surveys are of high precision and extend over large areas

PLANE SURVEYING:

The type of surveying in which the mean surface of the earth is considered as a plane, or in which its spheroidal shape is neglected, with regard to horizontal distances and directions

Ms .Azmah Marzia

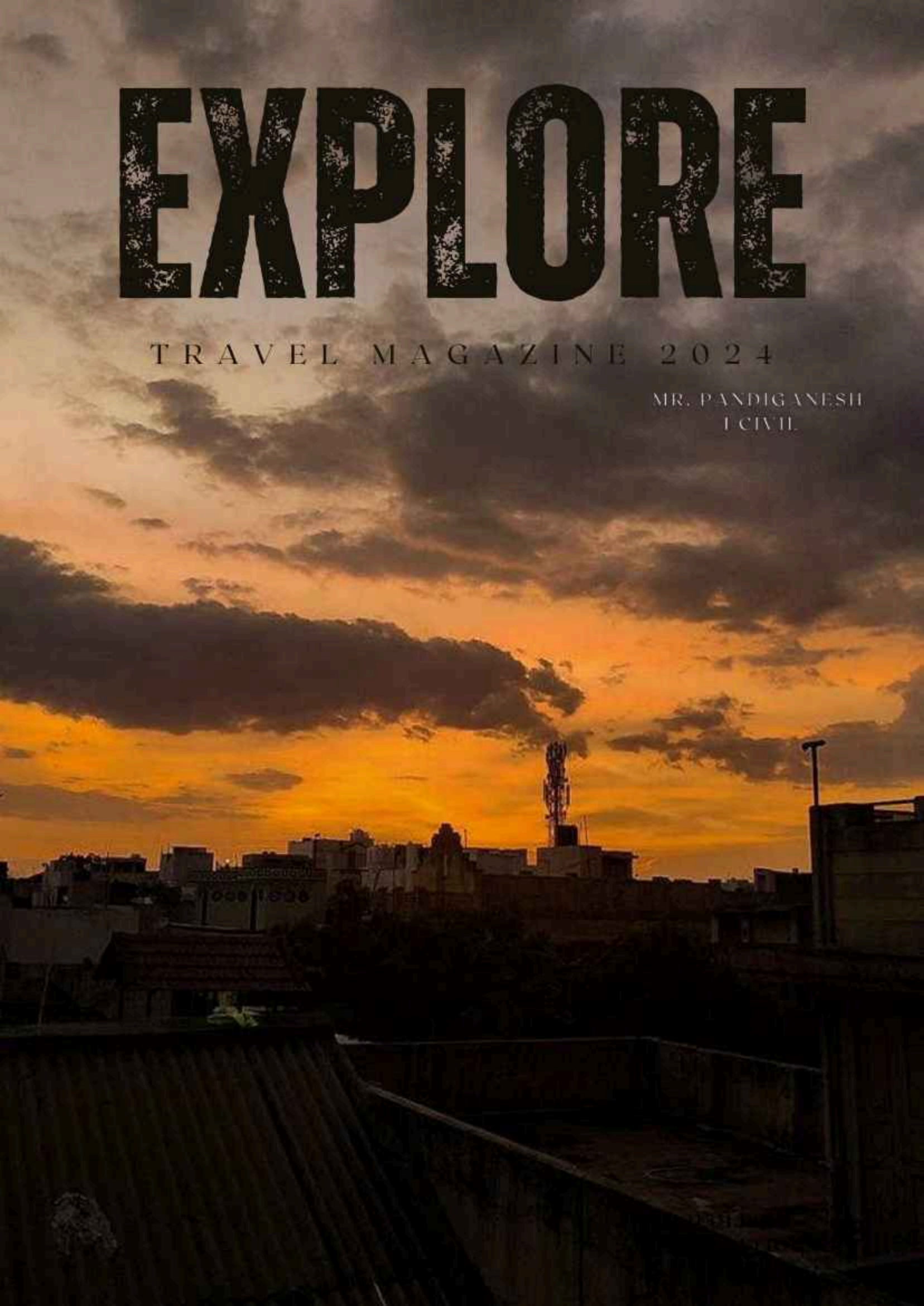
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PHOTOGRAPHY

EXPLORE

TRAVEL MAGAZINE 2024

MR. PANDIGANESH
I CIVIL



THANJAVUR

*The most
visited tourist
destination in
Tamilnadu*

*Sudharsan R
1st year*

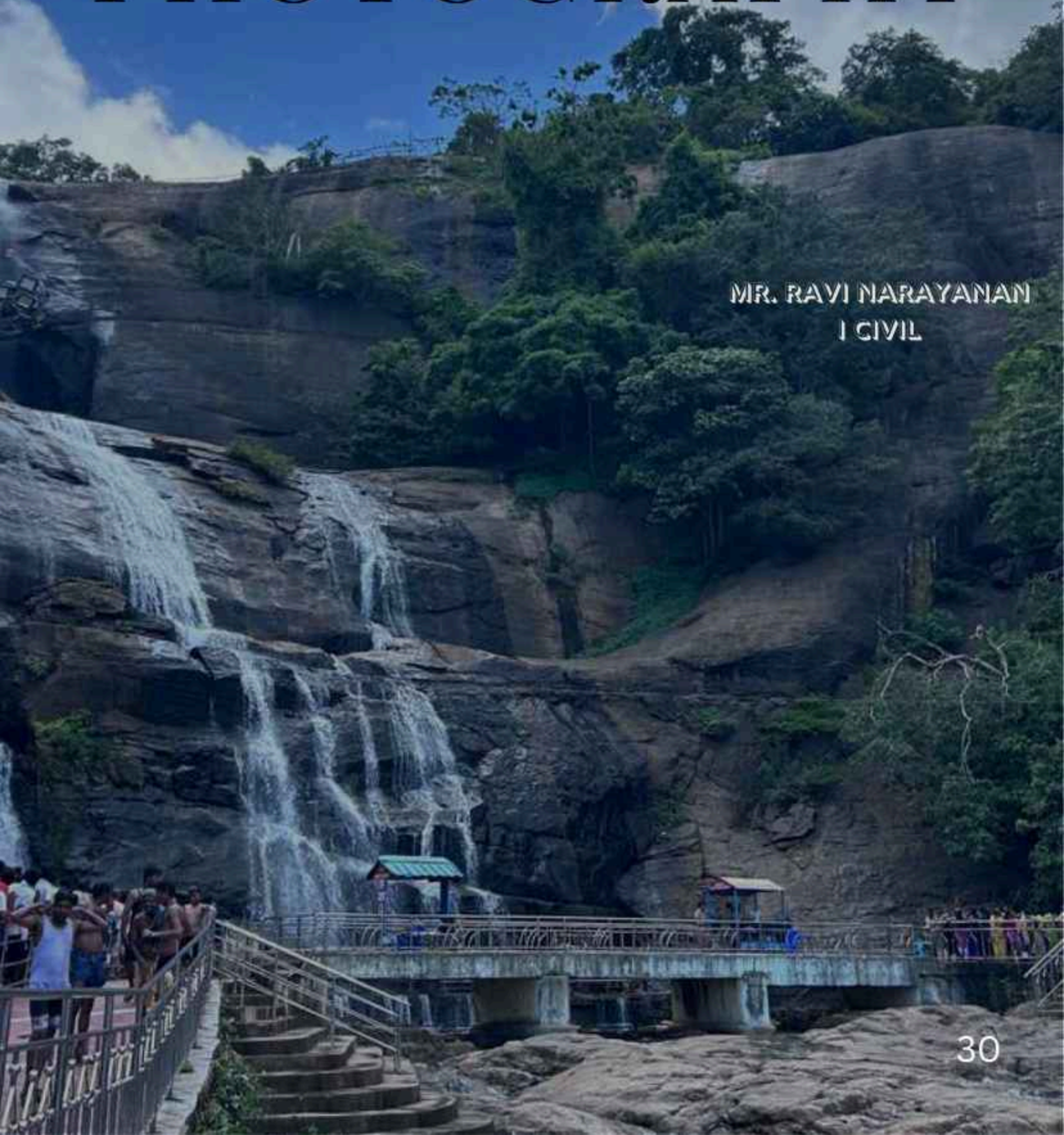
TRAVEL
EDITION



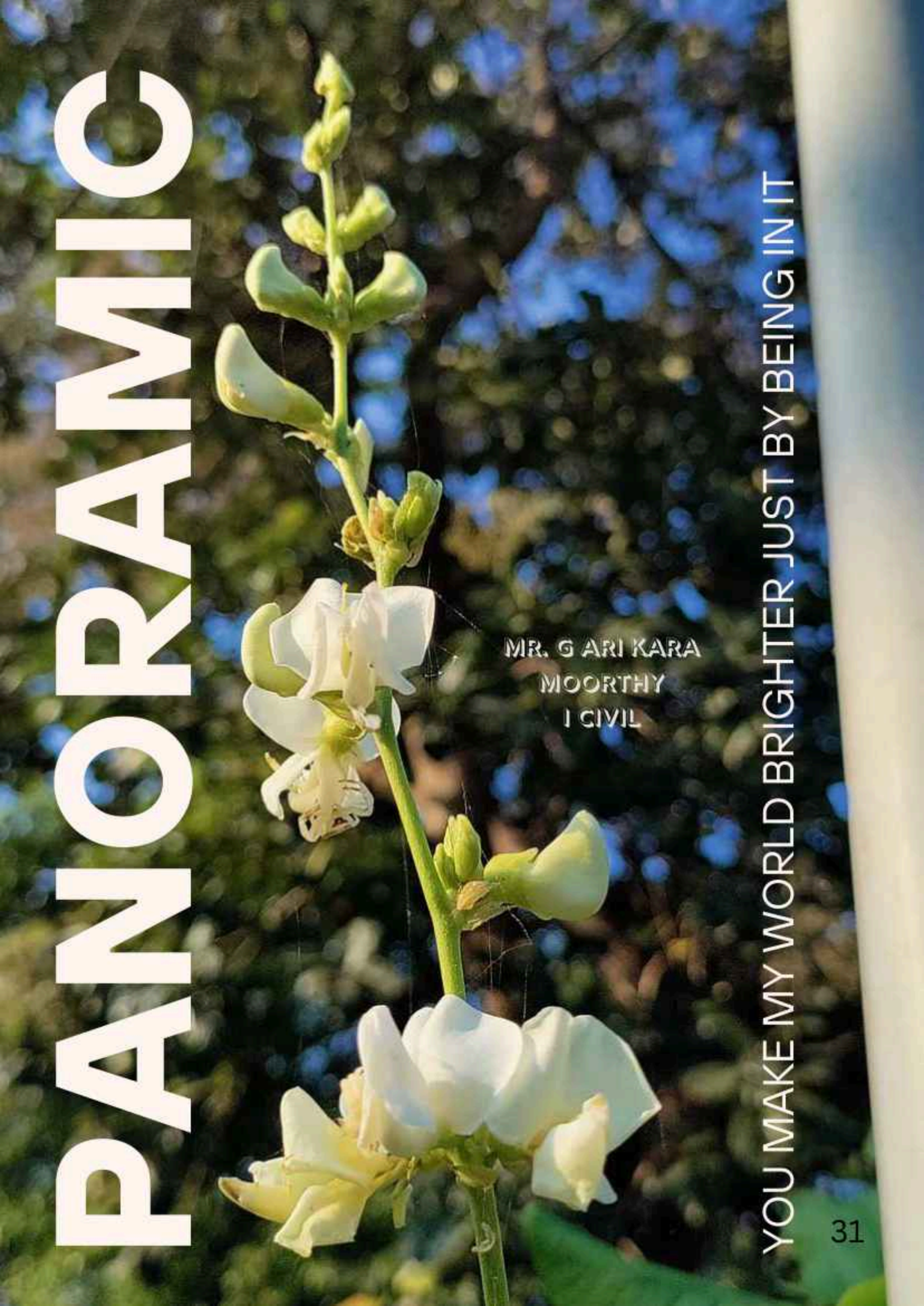


PHOTOGRAPHY

MR. RAVI NARAYANAN
I CIVIL



PANORAMIC



MR. G ARI KARA
MOORTHY
I CIVIL

YOU MAKE MY WORLD BRIGHTER JUST BY BEING IN IT

The background is a dark blue gradient. In the top right, there is a 3D wireframe cube with a cyan-to-magenta color gradient. In the top left, there is a 4x4 grid of white dots. In the bottom left, there is a glowing, multi-colored ring. In the bottom right, there is another 4x4 grid of white dots.

POSTER MAKING

Presented by civil

CLIMATE CHANGES

DON'T & DO

E. Sidhu
G. Ari Kara moorthy
S. Gopi kannan
From: B.E civil engineer
1st year

climate changing

Don't

1. Don't litter.
2. Don't cut trees.
3. Don't pollute the air.
4. Don't burn trees.
5. Don't waste the water.
6. avoid plastic bags.

Do's

1. Plant the trees.
2. turn off light.
3. reuse, reduce, recycle.
4. Use bicycles.
5. use hybrid cars.
6. Use solar energy.



WASTE MANAGEMENT AND OPTIMISATION



1

abstract



'Intelligent Waste Management Optimization' leverages IoT and data analytics to enhance sustainability. This innovative approach reduces waste collection costs by 25%, boosts recycling rates by 30%, and cuts emissions by 20%.

2

waste Generating



The world generates 2.01 billion tonnes of municipal solid waste annually, and is expected to grow to 3.40 billion tonnes by 2050. The cost of waste management in 2020 was estimated at USD 252 billion, but could increase to USD 640.3 billion by 2050.

3

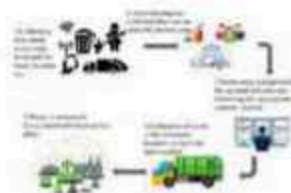
Roles of IOT



IoT devices and sensors can identify and sort different types of waste, such as plastics, metals, or organics, which can lead to more efficient recycling and reduce waste in landfills. It can enable real-time quality monitoring, which can help managers identify nonconformities and make process adjustments to reduce waste.

4

Method of usage



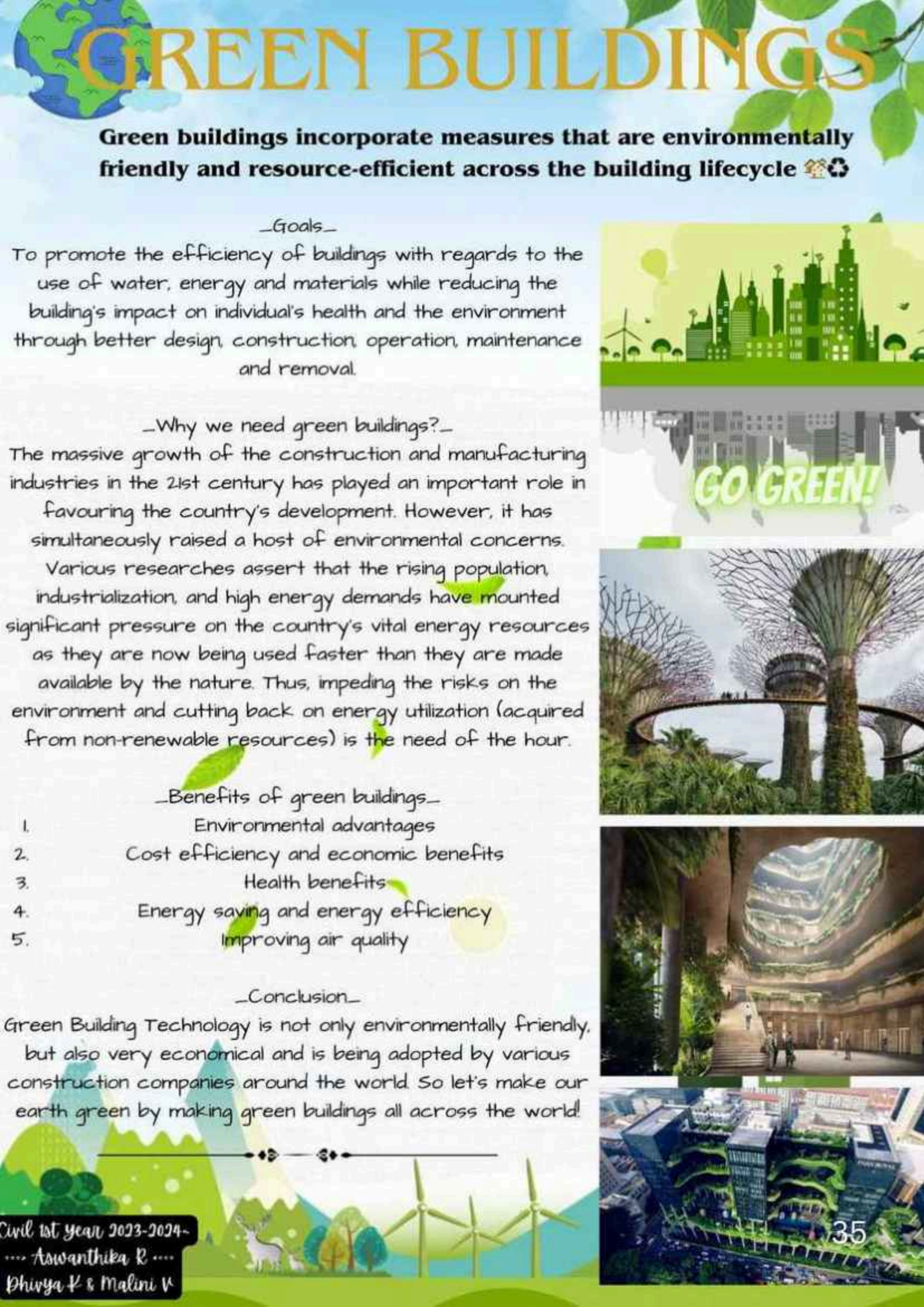
These innovative systems equip bins with sensors and communication modules, transforming them into a network of intelligent trash collectors. The heart of a garbage monitoring system lies in its sensors. These devices continuously gather data on the fill level within the bin. Ultrasonic sensors are a popular choice. They emit sound waves and measure the time it takes for the echo to return.

5

Advantages and outcomes

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waste management can reduce the environmental impact of waste management. For example, optimized collection routes can reduce the number of trucks on the road, which can reduce traffic congestion and carbon emissions. Sensors can detect toxic fumes or chemicals and provide real-time alerts of accidents or facility issues.



GREEN BUILDINGS

Green buildings incorporate measures that are environmentally friendly and resource-efficient across the building lifecycle ♻️

—Goals—

To promote the efficiency of buildings with regards to the use of water, energy and materials while reducing the building's impact on individual's health and the environment through better design, construction, operation, maintenance and removal.

—Why we need green buildings?—

The massive growth of the construction and manufacturing industries in the 21st century has played an important role in favouring the country's development. However, it has simultaneously raised a host of environmental concerns.

Various researches assert that the rising population, industrialization, and high energy demands have mounted significant pressure on the country's vital energy resources as they are now being used faster than they are made available by the nature. Thus, impeding the risks on the environment and cutting back on energy utilization (acquired from non-renewable resources) is the need of the hour.

—Benefits of green buildings—

1. Environmental advantages
2. Cost efficiency and economic benefits
3. Health benefits
4. Energy saving and energy efficiency
5. Improving air quality

—Conclusion—

Green Building Technology is not only environmentally friendly, but also very economical and is being adopted by various construction companies around the world. So let's make our earth green by making green buildings all across the world!



You are
NEVER too
small to
make a
difference



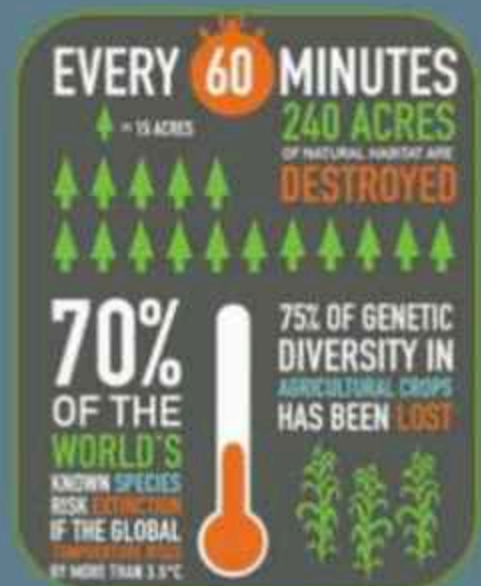
Sick of
pollution



TAKE
CARE
EARTH



'There is no
planet B'



HOW INTELLIGENT OF A
SPECIES ARE WE WHEN WE
DESTROY OUR ONLY HOME?



PROTECT
OUR
PLANET



**GLOBAL WARMING
ISN'T A
PREDICTION. IT IS
HAPPENING.**
- JAMES HANSEN



By:
Bala surya Kumari
Samyuktha
Jeevika Kannan



THANK YOU

ALL YOU NEED IS LOVE
